Ethan Gu

858-262-9018 ethangu4@berkeley.edu linkedin.com/in/ethan-gu4 github.com/EthanGu4

Education

University of California, Berkeley May 2027
B.S. in Electrical Engineering and Computer Science (EECS) Major GPA: 3.63/4.0
Coursework: Data Structures, Efficient Algorithms, Computer Architecture, Machine Structures, Database Systems, Web Development, Information Processing, Discrete Math, Probability Theory, Linear Algebra & Differential Equations

Technical Skills

Languages: Python, Java, Ruby, C, C++, HTML/CSS, JavaScript, TypeScript, SQL
Technologies: React, Ruby on Rails, Svelte, Next.js, PostgreSQL, NodeJS, Django, REST APIs, Git, AWS

Experience

Offerday AI July 2025 – Aug. 2025
Software Engineer Intern *San Diego, CA*

- Evaluated AWS Amplify and AWS Lambda as serverless alternatives to Amazon EC2
- Benchmarked performance, scalability, and operational cost under production workloads; implemented architecture improvements projected to reduce infrastructure spend by 22%
- Built a General Data Protection Regulation (GDPR) compliance assessment platform for secure data handling
- Automated conversion of regulatory requirements into a 13-point checklist, streamlining EU market deployment
- Audited backend systems for CCPA compliance by reviewing data flow architecture and storage practices; identified key engineering gaps and proposed changes to meet consumer privacy standards

Cal Blueprint Jan. 2025 – Present
Software Engineer *Berkeley, CA*

- Developed full-stack apps for nonprofit organizations serving users nationwide using modern web technologies
- Worked in an agile environment with designers and PMs to scope requirements and ship user-focused features
- Applied best practices in version control, code review, and testing to deliver maintainable, scalable software

Projects

Alliance for Clean Energy New York | *React Native, TypeScript, Supabase, Maps API*

- Mapped and modeled 130+ renewable energy projects statewide since 2017, permitting data-driven insights for policymakers, researchers, and the public to support clean energy goals
- Delivered visualizations covering 2.1 GW+ of clean energy to support NY's 70% by 2030 and 100% by 2040 goals

Amigos del los Rios | *React Native, TypeScript, Supabase, PlanIT Geo API*

- Developed web app for Emerald Necklace initiative, advancing urban greening and environmental justice efforts
- Seamlessly integrated PlanIT Geo API to ingest and visualize geospatial data for 10k+ trees and assets
- Optimized route workflows to minimize field time and maximize tree-watering coverage for volunteers

Blacktop Gems | *Ruby on Rails, Python, Hotwire, PostgreSQL, AWS RDS, NBA API*

- Built a full-stack analytics platform using NBA performance data for 530 actives via automated ingestion pipelines
- Designed a configurable player-ranking algorithm combining 8 weighted performance metrics and recent-game deltas, transforming raw data into normalized signals to surface statistically significant "hidden gem" players
- Implemented an automated sync and backfill pipeline supporting 10k+ database updates/day reliably

Fortuna Chess | *JavaScript, HTML/CSS*

- Built a turn-based, chess-inspired web game with 30+ custom pieces, terrain effects, and dynamic strategies
- Engineered a virtual game board model to decouple game logic from UI, supporting robust move validation, improving ability resolution speed by roughly 60%, and enabling 100% deterministic state testing
- Designed drag-and-drop interface and implemented event-driven DOM updates for responsive gameplay
- Composed 35+ tailored abilities across custom game pieces, improving replayability and strategic depth

PokéPilot | *React, Next.js, PostgreSQL, PokeAPI*

- Engineered a full-stack Pokémon team builder with JWT-based authentication and scalable user state management
- Integrated PokeAPI to fetch and normalize data for 1000+ Pokémon, powering interactive team-building workflows
- Designed secure REST APIs and database schemas to support 500+ teams, user sessions, and scalable features